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SIPLM: A language model for predicting human protein self interaction

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Abstract. Self-interacting proteins (SIPs) are crucial in cellular regulation, but experimental methods for their identification face limitations such as high costs, inefficiency, and false-positive rates. To address this, we are proposing a novel language model framework (SIPLM) that automatically predicts self interaction using protein sequences. Language model helps computers understand human language. It integrates local and global representations for end-to-end sentence processing. Applying Language model in computational biology has certain limitations which are addressed by ProteinBERT, a model specifically designed for protein sequences. Language models have never been applied to predicting protein self interactions. Moreover, self interaction data is not readily available, data was never published by previous researchers. Major contribution of this research is to prepare a human SIP dataset and examine if the language model can outperform state-of-the-arts. Dataset is prepared by collecting human SIP proteins from the latest version of protein-protein interaction databases. ProteinBERT is finetuned on this data to obtain proposed language model SIPLM. Cross validation AUC is 89% and AUC on held out set is 78%. Both of them are significantly higher than state-of-the-art methods. The dataset and the developed models are available at <https://github.com/CMATERJU-BIOINFO/SIPLM> for academic use only.

Keywords: Protein self interaction, Protein-protein interaction, SIP, PPI, Deep language model

1 Introduction

Cells, the fundamental structural and functional units of organisms, often contain thousands of proteins. Proteins, as essential cellular components, interact

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with other molecules to maintain normal physiological functions. Among these interactions, protein-protein interactions (PPIs) have been a central focus of biological research due to their significance in understanding cellular processes. A key subset of PPIs is self-interacting proteins (SIPs), which involve proteins that interact with identical copies of themselves, often encoded by the same gene. These interactions can lead to the formation of homo-oligomers, which play crucial roles in enzyme activation, signal transduction, immune responses, and gene expression regulation [6].

Studies have highlighted the importance of homo-oligomerization in various biological processes, [5] emphasizing its role in the evolution of cellular functions and protein interaction networks (PINs). This understanding aids in exploring cellular mechanisms systematically and provides a foundation for developing new drug targets and design strategies. Additionally, SIPs enhance protein stability by reducing surface area exposure, thereby preventing denaturation. Given these critical roles, there is an increasing need for efficient and reliable computational methods to complement traditional experimental approaches for identifying SIPs.

Numerous studies have been conducted to develop computational approaches for predicting protein-protein interactions (PPIs). For example, Zahiri et al. [12] developed PPIevo, a sequence-based model that uses position-specific scoring matrices to extract evolutionary information, demonstrating improved prediction accuracy. In recent times DensePPI [4] is published that uses a unique encoding strategy. Here rectangular images have been generated from two interacting(or non-interacting) proteins from color map which were then used to train DenseNet model to predict PPIs. Another work [8] in recent times uses deep neural network for PPI prediction

Despite these advancements, existing methods face limitations. Most PPI prediction models are not fully applicable to self-interacting proteins (SIPs), and their predictions for SIPs often lack effectiveness. Furthermore, the development of computational models for SIPs remains limited, with existing models frequently neglecting the challenges posed by unbalanced data sets. Addressing this issue is critical to improving prediction performance. Thus, exploring efficient and robust computational methods is important.

Modern deep neural network architectures, such as BERT [3], designed specifically for sequence data, coupled with pre-training on large datasets, have revolutionized automated text analysis. In particular, the attention-based Transformer architecture has demonstrated exceptional performance across a wide variety of benchmarks and domains.

Inspired by the remarkable success of BERT in natural language processing (NLP), it has recently been adapted for biological sequence analysis [2], [1]. Protein-BERT is a specialized adaptation of the BERT (Bidirectional Encoder Representation from Transformers) architecture, tailored for protein analysis. It leverages BERT's robust pre-training and transfer-learning capabilities to model the complex properties of proteins. The Protein-BERT architecture includes multiple layers of self-attention mechanisms and feed-forward neural

networks, similar to the original BERT framework. Its design features a unique encoding mechanism that captures long-term evolutionary patterns and relationships between amino acids. This bidirectional processing enables the model to consider both preceding and succeeding amino acids when encoding a specific position, making it highly effective for analyzing the intricate structures and interactions characteristic of many proteins.

In this study, we utilized deep language model for self interaction prediction. Language model is fine-tuned using experimental data related to self interactions, creating a refined model specifically designed for SIP prediction. The fine-tuning process involves training the model with human proteins taken from the latest version of BioGRID [9], DIP [11], STRING [10] and MINT [7].

The proposed method significantly improves classification performance than state-of-the-art methods. AUC is 86% which is higher than state of the art methods. For final classification, a one-star consensus strategy was employed, leading to a performance increase of over 4%. This highlights the effectiveness of the model in enhancing predictive accuracy and reliability.

2 Dataset

Self interaction data is not readily available as none of the previous researchers have published their. Also, previous works are based on old versions of the PPI datasets. So we have no other option than preparing the data from the existing PPI databases. We have chosen human proteins for our study. Human proteins have been taken from latest version of BioGRID [9], DIP [11], STRING [10] and MINT [7].

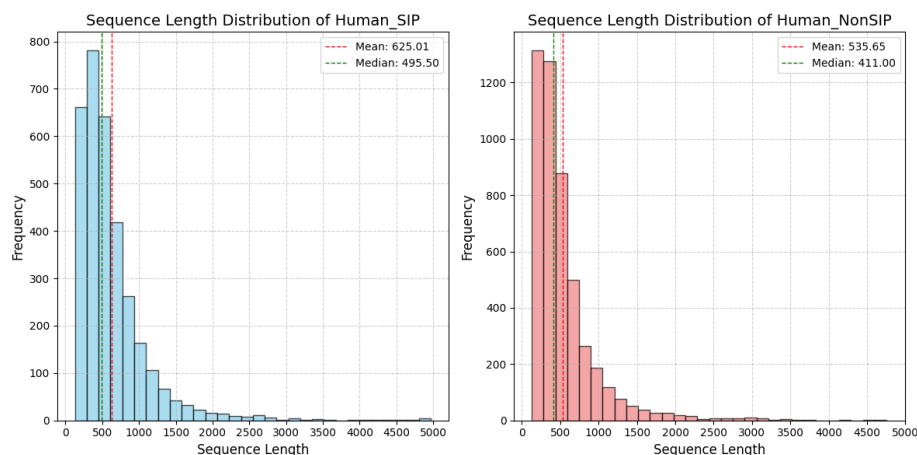


Fig. 1. Distribution of SIPs and non-SIPs, sequence length wise

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BioGRID (Biological General Repository for Interaction Datasets) is a free and open-access database that compiles and organizes information on protein, genetic, and chemical interactions. It encompasses interaction data for many organisms including human.

The Database of Interacting Proteins (DIP) is a biological resource that compiles experimentally verified protein-protein interactions. By integrating data from various sources, DIP provides a unified and consistent set of interaction records. The database is curated through a combination of manual review by expert curators and automated computational methods.

3 Methodology

3.1 Data preparation

Data preparation steps have been explained in Table 1:

Table 1. Human self interaction data preparation steps:

Data preparation steps:	
1	Total 16414 unique human proteins were found in Uniprot database.
2	Self interactions mean protein ID's of two interacting proteins are same.
3	Only proteins with sequence length between 128 and 5000 are considered.
4	Proteins, with 80% similarity, have been removed using cd-hit technique.
5	Count of SIPs are less than non-SIPs. However, average sequence length of SIPs are higher than that of non-SIPs. Subset of on-SIPs are selected randomly to maintain class balance.
6	Final dataset contains 3288 SIPs and 4847 non-SIPs.

Dataset so prepared is divided into two part : first part is further divided into 10-fold for using it in cross validation. Second part is kept as held out set. Train test set-up is explained in workflow diagram 2.

3.2 Model Selection

A deep language model has been employed for classification tasks, with its architecture drawing inspiration from BERT (Bidirectional Encoder Representations from Transformers), developed by Google AI in 2018. Since its introduction, BERT has revolutionized the field of NLP with its exceptional performance across various language understanding tasks. As a transformer-based model pre-trained on a vast corpus of text data, BERT is capable of capturing intricate linguistic patterns and developing contextual language representations.

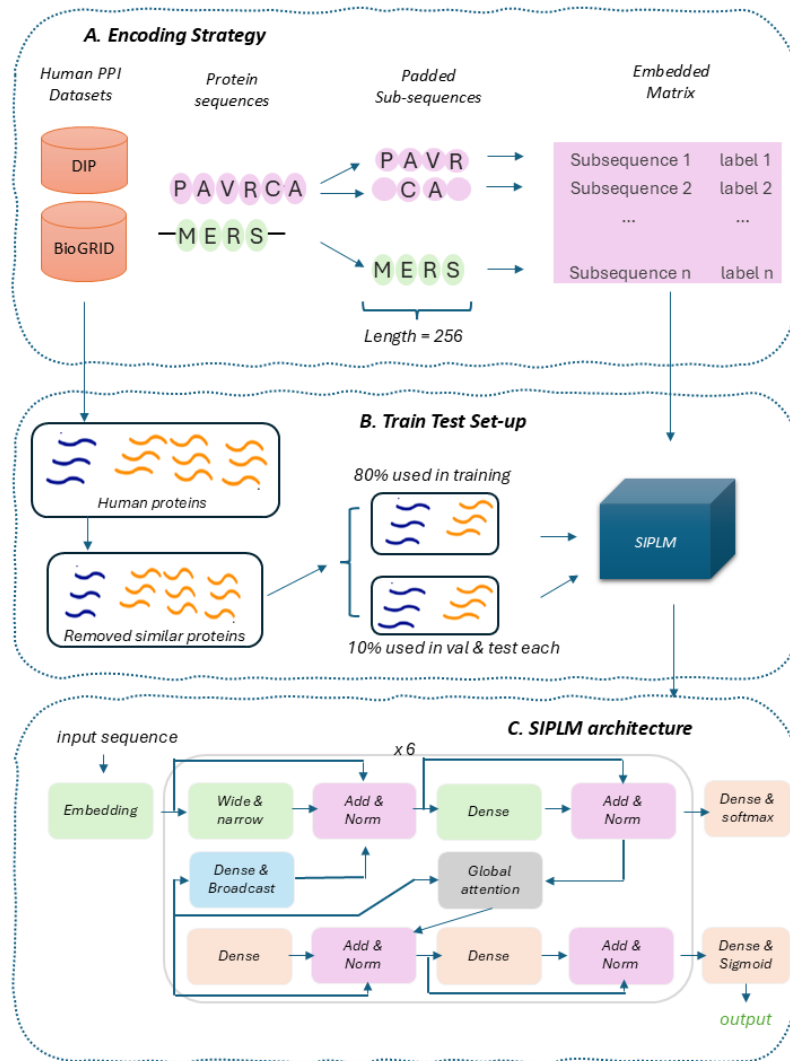


Fig. 2. Overall block diagram of this experiment: A. First, protein sequences were encoded and converted to embedded matrices as per standard input for language model. B. In this experiment, human proteins from BioGRID and DIP have been used. Proteins were identified as SIP or not, similar (80%) proteins discarded. Then 80% of the data was used in training and remaining for test and validation. C. SIPLM architecture: Finetuned ProteinBERT [1] is used for prediction.

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In recent years, researchers have explored BERT's potential in diverse domains, including bioinformatics and computational biology, extending its applications far beyond traditional NLP tasks. One particularly promising area of application is the analysis of protein sequences.

Proteins, as essential biological molecules, play a critical role in drug discovery and other scientific fields, making the understanding of their structure and function a priority. BERT's adaptability has made it a valuable tool in advancing research in this domain.

3.3 Training Process

During fine-tuning, protein sequences and their corresponding ground truth values are fed into the model. Proposed model accepts sequences of length less than 512. SIPLM uses positional encoding to convert protein sub-sequences in to embedded matrices. Encoding strategy is explained in workflow diagram 2. Unique tokens of amino acids along with additional tokens (<START>, <END>, etc) have been used for encoding. Shorter sequences are padded with <PAD> tokens on both side. Embedded matrix passed into SIPLM model as input. Learning rate is reduced if performance is not improved over epochs.

3.4 Test Strategy

A separate dataset (i.e. held out data) has been used for comparing performance with state of the art methods.

Following two test strategies have been used in this paper:

1. **Average** : Held out data were fed into ten generated models to get ten prediction results. Then average performance is reported.

$$FC_k = \begin{cases} 1, & \text{if } count_k(1) \geq count_k(0) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

2. **Consensus Strategy**: 1-hot rule has been used. Held out set given to ten models to get five classification results. Final prediction result is generated using 1-hot rule FC_k .

4 Results and discussions

Our proposed method is able to predict self interaction with high accuracy. Table 2 shows 10-fold cross validation performance.

Performance of the proposed SIPLM model is compared with other state of the art methods on held out dataset. With respect to the method of Li et al.[6], the performance has been considered with our proposed method. The performance enhancements has been shown in Table 3.

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Table 2. 10 fold cross validation result on human SIP dataset using proposed method i.e. SIPLM

Fold	Accuracy	AUC	Precision	Recall	F1-score
1	0.8034	0.8852	0.7690	0.8095	0.7887
2	0.8019	0.8892	0.7865	0.7843	0.7854
3	0.7973	0.8885	0.7598	0.7598	0.7886
4	0.8012	0.8926	0.7651	0.8161	0.7898
5	0.8038	0.8964	0.7624	0.8315	0.7954
6	0.7973	0.8885	0.7615	0.8068	0.7835
7	0.8344	0.9235	0.7939	0.8640	0.8275
8	0.8098	0.8954	0.8242	0.7458	0.7830
9	0.8049	0.8922	0.7884	0.7848	0.7866
10	0.8441	0.9254	0.8318	0.8250	0.8284

For the final classification, a one-hot consensus strategy was employed. Consensus approaches aim to aggregate and integrate data from multiple sources to enhance reliability and overcome the limitations of individual methods. This strategy improved classification performance compared to standalone models. As shown in Table 3, the proposed method outperforms previously established state-of-the-art techniques.

Table 3. Performance comparison on held-out data (test set)

Method	Accuracy	AUC	Precision	Recall	F1-score
Li et al [6]	62	66	60	62	58
SIPLM (proposed)	73	78	65	70	67

Confusion matrix of 1* hot rule is given in figure 3 shown at protein level model performance.

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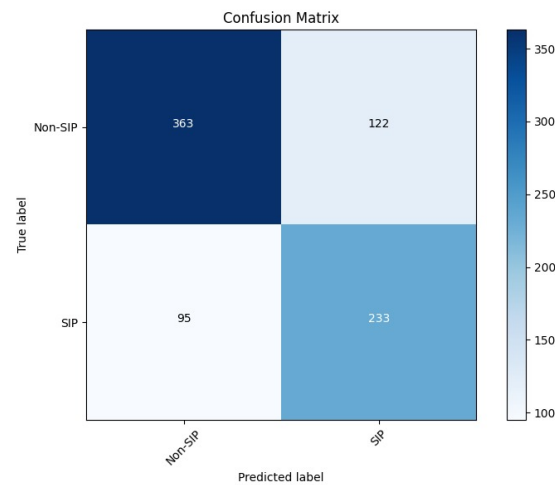


Fig. 3. Confusion matrix for classifying into SIP and non-SIP at protein level

5 Conclusion

As part of this research work, we made the human protein Self interaction data readily available, which was not available earlier. A fine-tuned version of the language model is proposed for predicting self interaction that outperforms all previous approaches. Language model has shown impressive results in other areas but have never been used for predicting protein self interaction. Our research indicates that performance improvement happening when a one-hot rule is used on sub-sequences to predict protein level classification.

In future, we are planning to work on other organisms including yeast and bacterial proteins. Other methods need to be explored as well.

6 Acknowledgment

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Conflicts of interests: The authors state that there are no conflicts of interest.

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